# Active Maintenance Report: coppock_etal_2017

**Coppock, Gerber, Green and Kern (2017, Political Analysis): Combining Double Sampling and Bounds to Address Nonignorable Missing Outcomes in Randomized Experiments**

2026-08-01

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**Archived at Zenodo:** doi:10.5281/zenodo.24168330. This DOI always resolves to the latest release of the maintained version.

*Drafted by Claude Opus 5 under the supervision of Alex Coppock.*

This repository holds the actively maintained replication code for Coppock, Gerber, Green and Kern (2017), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

| | |
|---|---|
| Article | 10.1017/pan.2016.6 |
| Replication archive | 10.7910/DVN/AQB4MP |
| Preanalysis plan | osf.io/wkj8v |

**The data are not redistributed here.** The deposit is 231 KB across seven files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

**Repository layout.** `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the hand extraction of the article, `build_ground_truth.R` generates the comparison and runs the coverage gate, and `coppock_etal_2017_ground_truth.csv` is what it writes, with `float_coverage.csv` beside it. `run_archive.R` runs the deposited scripts from a scratch copy of the deposit and writes `archive_run_status.csv` and `archive_overwrites.csv`; it installs the deposit's own packages and is run by hand rather than from `run_all.R`. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` corrects the published article, rendering to `coppock_etal_2017_errata.pdf` at the root. This file is the reproducibility report, also available as a PDF in `report/`.

**License.** CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

**To reproduce.** Clone or download the repository, open `coppock_etal_2017.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its seven files, installs the `attrition` package from the deposited source tarball if it is not already present, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, modelsummary, knitr, kableExtra, gridExtra, here, and `attrition`, which `install_attrition.R` handles. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about seven seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The three PDF figures differ on every run because a PDF records the time it was written; the CSV, TeX and PNG output comes back byte-identical.

# 1 Summary

Two questions, answered before the detail.

## 1.1 Does the deposited archive run?

Not without intervention, and the obstacle is a bug in the deposit rather than anything that has aged. `experimental_analysis_main_text.R` prints Tables 1, 2 and 3 and then stops at a line that subtracts rows from an object called `mturk_results`, which the script never creates. The object it means is `table_3`, assembled four lines earlier under a different name. Everything after that line, which is the 65 percent confidence interval reduction the results section quotes and the entire sensitivity analysis behind Figure 2, is unreachable. That leaves 3 recorded claims with no archive value at all, all of them numbers the article prints: the reduction in interval width, the sensitivity parameter as Section 4.3 states it, and the same parameter as Figure 2 prints it on its face.

The other two scripts run. `simulation_analysis.R` produces Figure 1 and Appendix Figure 1 with two deprecation warnings and no errors. `experimental_analysis_appendix.R` runs and reproduces Appendix Table 1 exactly, but only against the `commarobust` package as deposited. The current version of `commarobust` on GitHub also calls itself 0.1.0 and exports nothing at all: it has been reduced to a message telling the user to install `estimatr` instead. A reader who follows the archive's own README, which recommends `devtools::install_github()` over the bundled tarball, gets a package that fails on the first line that uses it, and `packageVersion()` reports the same number either way.

The remaining 128 claims the archive can be compared on all reproduce, including every cell of all five published tables.

### 1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. Of the 171 recorded claims, 158 carry a published number and all 158 match it to the precision the page prints; 0 differ. The remaining 13 are claims about shape rather than value, and all 13 hold. The three the archive cannot reach are among the matches: replacing the stale object name gives a 65.0 percent reduction in confidence interval width where the paper says 65 percent, and the sensitivity analysis gives a $\delta^*$ of 0.0707 where the paper and Figure 2 both say 0.07.

No number in the article is contradicted by the data behind it. A word-by-word reading of the article and its appendix did find errors of another kind, in the prose, the citations, and one table's headings; none changes an estimate or a conclusion, and they are set out in `coppock_etal_2017_errata.pdf` (see Errata).

## 2 Paper overview

**Citation**: Coppock, A., Gerber, A. S., Green, D. P. and Kern, H. L. (2017). "Combining double sampling and bounds to address nonignorable missing outcomes in randomized experiments." *Political Analysis*, 25(2), 188-206. DOI: 10.1017/pan.2016.6

**Summary**: Missing outcomes break the guarantee randomization provides, and the usual repairs assume that missingness is ignorable. The paper proposes an alternative that does not: combine a double sampling design, in which a random subset of the initial nonrespondents is pursued with a larger incentive, with nonparametric worst-case bounds on the outcomes still missing after that second effort. It derives the bounds estimator under double sampling, gives analytic variances and confidence intervals, adds poststratification for covariate adjustment, and adds a sensitivity analysis indexed by the fraction $\delta$ of remaining nonrespondents for whom ignorability fails. Monte Carlo simulations show how the identification region and the confidence interval respond to the two response rates. The application is a two-wave Mechanical Turk replication of Levendusky and Malhotra (2016) in which 1,980 subjects read either a polarized or a moderate news story, 1,444 responded in the initial Wave 2 sample, and 50 nonrespondents per arm were offered four times the original incentive, of whom 72 complied. Worst-case bounds from the initial sample alone are $(-1.54, 1.71)$ and say nothing. Double sampling with poststratification narrows the associated confidence interval from 3.50 to 1.23, a 65 percent reduction, while the sensitivity analysis shows the significant naive estimate collapses once ignorability fails for even 7 percent of the remaining nonrespondents.

## 3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

| Script | Status on current R | Resolution |
|---|---|---|
| experimental_analysis_main_text.R | Prints Tables 1, 2 and 3, then stops: object 'mturk_results' not found | The object meant is table_3, built four lines above under another name |
| experimental_analysis_appendix.R | Runs against the deposited commarobust; fails against the current one | install.packages('original/commarobust_0.1.0.tar.gz', repos = NULL, type = 'source') |
| simulation_analysis.R | Runs; two deprecation warnings | guide = FALSE becomes guide = 'none'; %+% becomes + |

The deposit is unusually well prepared in one respect and unusually exposed in another. It bundles the source tarballs of both packages the analysis depends on, `attrition` and `commarobust`, neither of which has ever been on CRAN. That is the right instinct, and it is what makes the appendix script reproducible today. The README then undercuts it: it recommends installing from GitHub instead, calling that route "easy," and offers the tarballs only "in the interest of full replicability." Ten years on, the easy route is the broken one.

## 3.1 The two packages

`attrition` implements the paper's method and is still on GitHub. The deposited version is 0.0.0.9000 and the current one is 1.0.0, and the two no longer agree everywhere. 1.0.0 fixed the Imbens-Manski critical value, which had been found by minimising an absolute value over a fixed interval that only covers alpha near 0.05, so the interval bounds move by about 2e-09 relative here and would move much more at another significance level. Every point estimate is unchanged. 1.0.0 also renamed the six quantities its estimators return, onto broom's names, and reports the uncertainty as a standard error where 0.0.0.9000 reported a variance. The maintained rewrite is written against the current package and `install_attrition.R` halts if an older one is installed, since the old names would otherwise fill these tables with `NA` rather than fail. The deposited tarball is still what the scripts in `original/` are written against, and it still runs them; `install_attrition.R` installs it when `attrition` is absent, because the tarball's checksum is pinned in `original_manifest.csv` and a GitHub repository's contents are not.

`commarobust` is a different case. It was the author's prototype for robust standard errors around `lm()`, and it was superseded by `estimatr`, whose `lm_robust()` defaults to the same HC2 estimator. The current GitHub `commarobust` has been emptied out to a startup message pointing at `estimatr`, and it exports no functions. It nevertheless still calls itself version 0.1.0, exactly like the deposited tarball, which exports six. Two packages with the same name and the same version number differ in whether the archive runs, and nothing in `sessionInfo()` would tell a reader which one they have.

### 3.2 The stale object name

The main text script builds Table 3 and then reaches for it under a name it does not have:

```
table_3 <- cbind(no_ds = cis_1, ds = cis_2, ds_ps = cis_3)
# ... four lines of row labels and a print ...
widths <- mturk_results[2,] - mturk_results[1,]
(widths[1] - widths[3]) / widths[1]
```

`mturk_results` is never assigned anywhere in the archive. The name presumably survived an interactive session in which the object had been called that, and the rename was not carried through before the file was deposited. This is the class of defect that only a clean-session run finds: in the author's own session the old object was still in the workspace, so the script worked.

The cost is not cosmetic. The line that fails computes the 65 percent reduction in confidence interval width, which is the sentence the results section builds toward, and everything below it, including the entire sensitivity analysis and Figure 2, never executes. Both values are correct once the name is fixed, so the archive contains a correct analysis that it cannot run.

## 4 Errata

The maintained rewrite makes one correction to the archive's code: `mturk_results` becomes the object that was meant. Nothing else in the estimator, the specification or the sample is changed.

Table 2: The published claims the deposited script cannot reach, and what the rewrite produces.

| Claim | Published | Archive | Rewrite |
|---|---|---|---|
| Section 4.2 reduction in 95% CI width | 65 percent | not computed, script stops | 0.6503 |
| Section 4.3 sensitivity parameter | 0.07 | not computed, script stops | 0.0707 |
| Figure 2 annotation | 0.07 | not computed, script stops | 0.0707 |

No published value changes. The correction recovers numbers the archive was unable to produce; all of them agree with the article. That is a correction to the deposited code, which is what this section is for. It is not an erratum in the article: every number the article prints is reproduced here, and every claim it makes about shape holds.

The article's own errata are set out in `coppock_etal_2017_errata.pdf` at the root of this repository: 11 corrections and 2 item that flags a probable error without correcting it. They come from reading the article and its appendix word by word, and none touches a printed number. Section 4.1 gives the size of the Moderate and Polarized arms (1,980) as the size of

the experiment, which randomized 2,955 subjects to three arms, and Table 1's headings place Responded and Did Not Respond under the follow-up sample. Four citations are wrong: An et al. (2009) is cited by its authors' given names, and Lee, Zhang and Rubin, and Si, Reiter, and Hillygus carry the wrong year. The rest are misspellings and dropped or run-together words, and the item left uncorrected is a probable stray word in a treatment stimulus whose fielded text we could not check.

## 5 The extraction and the two instruments

Two files stand between the article and this pipeline, and they are deliberately separate.

`ground_truth/published_claims.csv` is the **extraction**: every numeric token in the article and the online appendix, read out by hand and classified. It carries 212 rows. Citation years, equation numbers, footnote markers, section cross-references and page numbers are excluded as typography rather than claims; everything else is here, including numbers spelled out in words ("three scenarios", "two decades", "the four policy questions") that no token scan finds. `value_paper` is the string the page prints, with its digits intact, and `digits` records that precision for each claim separately, because the appendix regression table prints one R-squared at three decimals and its neighbour at four.

Table 3: The extraction, by claim type.

| Claim type | Needs a claim block | Rows |
|---|---|---|
| definitional | no | 26 |
| definitional | yes | 16 |
| descriptive | yes | 13 |
| pipeline | yes | 138 |
| structural | yes | 4 |
| transcribed | no | 15 |

A block is required for 171 of those rows. The 41 that do not need one are quantities the deposit was never going to record and that no code can therefore check: the $1.00 and $4.00 payments, the ten days between waves, the seven-point item scales, and the percentages inside the treatment articles themselves, which are quoted from the source study. They are verified against the article's own methods section and at the point of use in the scripts, and they are named here so that a reader can see the boundary rather than infer it. The simulation's outcome means and variances are in the same class: they are inputs to `attrition:::ds_manski_cis_2s()` that no output file carries, and they sit beside the call in `figure_1_simulation.R`.

`maintained/in_text_claims.R` is the **second instrument**. It reads only `maintained/output/`, never the ground truth, and it recomputes each claimed number by its own path: the prose figures of Section 4.2 are derived there from `table_3_bounds_main.csv` and

`table_a1_mipo_effects.csv` rather than read out of `text_section_4_claims.csv`, which is the file the ground truth reads. It prints one line per claim, and `ground_truth/build_ground_truth.R` runs it as a program, in its own environment, and asserts three things: that the number of claims printed equals the number the extraction requires, that the two sets of identifiers are equal in both directions, and that every printed value equals the ground truth's own value rendered at the same precision.

Two of the article's claims are checkable only because a second quantity can be recovered from the tables the article prints. Section 4.2 says missing outcomes are replaced with the lowest possible value of 0 and the highest of 6. Neither limit appears in any output file, but with the response rates of Table 1 and the initial-responder means of Table 2, the worst-case low and high estimates of Table 3 are two linear equations in exactly those two limits, and solving them returns 0 and 6.

Coverage of the published floats is complete for the tables: 15 of 15 cells in Table 1, 12 of 12 in Table 2, 18 of 18 in Table 3, 32 of 32 in Appendix Table 1 and 36 of 36 in Appendix Table 2, or 113 published cells with 113 rows. The three figures print no estimates except the $\delta^* = 0.07$ that Figure 2 carries on its face, which is covered. What the figures do assert is shape, and those assertions are covered by claims with computed truth values rather than by cells: that the identification region collapses to a point when everyone responds to the follow-up, that at a follow-up response rate of zero its width returns to the worst-case width of $2(1 - p_1)$, that it narrows as the initial response rate rises, and that raising the sample size a hundredfold shrinks the confidence interval while leaving the identification region unchanged to the last decimal.

## 6 Number-by-number comparison

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured.

| Location | Claim | Paper | Archive | Match |
|---|---|---|---|---|
| Appendix Figure 1 | appx_ci_shrinks_region_does_not | | | |
| Appendix Table 1 | ta1_all_ex_wave1_n | 2156 | 2156 | TRUE |
| Appendix Table 1 | ta1_all_ex_wave1_r2 | 0.001 | 0.001 | TRUE |
| Appendix Table 1 | ta1_all_ex_wave2_n | 2156 | 2156 | TRUE |
| Appendix Table 1 | ta1_all_ex_wave2_r2 | 0.0004 | 0.0004 | TRUE |
| Appendix Table 1 | ta1_all_pp_wave1_n | 2156 | 2156 | TRUE |
| Appendix Table 1 | ta1_all_pp_wave1_r2 | 0.024 | 0.024 | TRUE |
| Appendix Table 1 | ta1_all_pp_wave2_n | 2156 | 2156 | TRUE |
| Appendix Table 1 | ta1_all_pp_wave2_r2 | 0.002 | 0.002 | TRUE |
| Appendix Table 1 | ta1_constant_ex_wave1_coef | 1.531 | 1.531 | TRUE |
| Appendix Table 1 | ta1_constant_ex_wave1_se | 0.024 | 0.024 | TRUE |
| Appendix Table 1 | ta1_constant_ex_wave2_coef | 1.617 | 1.617 | TRUE |
| Appendix Table 1 | ta1_constant_ex_wave2_se | 0.024 | 0.024 | TRUE |
| Appendix Table 1 | ta1_constant_pp_wave1_coef | 3.108 | 3.108 | TRUE |
| Appendix Table 1 | ta1_constant_pp_wave1_se | 0.046 | 0.046 | TRUE |
| Appendix Table 1 | ta1_constant_pp_wave2_coef | 3.542 | 3.542 | TRUE |
| Appendix Table 1 | ta1_constant_pp_wave2_se | 0.046 | 0.046 | TRUE |
| Appendix Table 1 | ta1_placebo_ex_wave1_coef | 0.019 | 0.019 | TRUE |
| Appendix Table 1 | ta1_placebo_ex_wave1_se | 0.035 | 0.035 | TRUE |
| Appendix Table 1 | ta1_placebo_ex_wave2_coef | -0.011 | -0.011 | TRUE |

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured. *(continued)*

| Location | Claim | Paper | Archive | Match |
|---|---|---|---|---|
| Appendix Table 1 | ta1_placebo_ex_wave2_se | 0.034 | 0.034 | TRUE |
| Appendix Table 1 | ta1_placebo_pp_wave1_coef | 0.352 | 0.352 | TRUE |
| Appendix Table 1 | ta1_placebo_pp_wave1_se | 0.067 | 0.067 | TRUE |
| Appendix Table 1 | ta1_placebo_pp_wave2_coef | 0.042 | 0.042 | TRUE |
| Appendix Table 1 | ta1_placebo_pp_wave2_se | 0.066 | 0.066 | TRUE |
| Appendix Table 1 | ta1_polarized_ex_wave1_coef | -0.031 | -0.031 | TRUE |
| Appendix Table 1 | ta1_polarized_ex_wave1_se | 0.034 | 0.034 | TRUE |
| Appendix Table 1 | ta1_polarized_ex_wave2_coef | -0.030 | -0.030 | TRUE |
| Appendix Table 1 | ta1_polarized_ex_wave2_se | 0.034 | 0.034 | TRUE |
| Appendix Table 1 | ta1_polarized_pp_wave1_coef | 0.468 | 0.468 | TRUE |
| Appendix Table 1 | ta1_polarized_pp_wave1_se | 0.067 | 0.067 | TRUE |
| Appendix Table 1 | ta1_polarized_pp_wave2_coef | 0.126 | 0.126 | TRUE |
| Appendix Table 1 | ta1_polarized_pp_wave2_se | 0.066 | 0.066 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_ci_lower | -0.4778 | -0.4778 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_ci_upper | 0.5769 | 0.5769 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_var_high | 0.0067 | 0.0067 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_var_low | 0.0083 | 0.0083 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_wcb_high | 0.4426 | 0.4426 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_wcb_low | -0.3284 | -0.3284 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_ci_lower | -0.6119 | -0.6119 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_ci_upper | 0.5044 | 0.5044 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_var_high | 0.0114 | 0.0114 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_var_low | 0.0106 | 0.0106 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_wcb_high | 0.3286 | 0.3286 | TRUE |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_wcb_low | -0.4424 | -0.4424 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_ci_lower | -0.6087 | -0.6087 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_ci_upper | 0.6032 | 0.6032 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_var_high | 0.0095 | 0.0095 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_var_low | 0.0070 | 0.0070 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_wcb_high | 0.4426 | 0.4426 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_ex_wcb_low | -0.4710 | -0.4710 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_ci_lower | -0.5283 | -0.5283 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_ci_upper | 0.7452 | 0.7452 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_var_high | 0.0111 | 0.0111 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_var_low | 0.0129 | 0.0129 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_wcb_high | 0.5718 | 0.5718 | TRUE |
| Appendix Table 2 | ta2_pol_vs_mod_pp_wcb_low | -0.3417 | -0.3417 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_ci_lower | -0.6004 | -0.6004 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_ci_upper | 0.6944 | 0.6944 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_var_high | 0.0094 | 0.0094 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_var_low | 0.0085 | 0.0085 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_wcb_high | 0.5350 | 0.5350 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_wcb_low | -0.4491 | -0.4491 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_ci_lower | -0.6171 | -0.6171 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_ci_upper | 0.7262 | 0.7262 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_var_high | 0.0114 | 0.0114 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_var_low | 0.0124 | 0.0124 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_wcb_high | 0.5502 | 0.5502 | TRUE |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_wcb_low | -0.4339 | -0.4339 | TRUE |
| appendix_section_1_p2 | appx_n_large | 80000 | | |
| appendix_section_1_p2 | appx_n_small | 800 | | |
| appendix_section_1_p2 | appx_n2_large | 5000 | | |
| appendix_section_1_p2 | appx_n2_small | 50 | | |
| appendix_section_1_p2 | appx_size_factor | 100 | | |
| appendix_section_1_p2 | appx_two_scenarios | 2 | | |
| appendix_section_2_p3 | appx_effect_diminished | | | |
| appendix_section_2_p3 | appx_extremity_negligible | | | |
| appendix_section_2_p3 | appx_wave1_effect | 0.468 | | |
| appendix_section_2_p3 | appx_wave1_effect_strong | | | |
| appendix_section_2_p3 | appx_wave2_effect | 0.126 | | |
| appendix_section_3_p4 | appx_pairwise_columns | 6 | | |
| Figure 1 | sim_p2_zero_is_wcb | | | |
| Figure 1 | sim_point_identified | | | |
| Figure 1 | sim_region_smaller_in_p1 | | | |

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured. *(continued)*

| Location | Claim | Paper | Archive | Match |
|---|---|---|---|---|
| Figure 1 | sim_width_shrinks_to_zero | | | |
| Figure 2 | fig2_delta_star | 0.07 | | FALSE |
| section_3_p11 | sim_n_followup | 50 | | |
| section_3_p11 | sim_n_total | 800 | | |
| section_3_p11 | sim_p1_max | 0.75 | | |
| section_3_p11 | sim_p1_min | 0.25 | | |
| section_3_p11 | sim_p2_max | 1 | | |
| section_3_p11 | sim_p2_min | 0 | | |
| section_3_p11 | sim_true_ate | 0 | | |
| section_3_p12 | sim_n_scenarios | 3 | | |
| section_3_p12 | sim_p1_scenario_high | 0.75 | | |
| section_3_p12 | sim_p1_scenario_low | 0.25 | | |
| section_3_p12 | sim_p1_scenario_mid | 0.50 | | |
| section_4.1_p14 | app_n_completed | 72 | | |
| section_4.1_p14 | app_n_invited | 100 | | |
| section_4.1_p14 | app_n_invited_per_arm | 50 | | |
| section_4.1_p14 | app_n_responded | 1444 | | |
| section_4.1_p14 | app_n_subjects | 1980 | | |
| section_4.2_p14 | app_outcome_max | 6 | | |
| section_4.2_p14 | app_outcome_min | 0 | | |
| section_4.2_p14 | txt_ci_width_ds_ps | 1.23 | 1.23 | TRUE |
| section_4.2_p14 | txt_ci_width_no_ds | 3.50 | 3.50 | TRUE |
| section_4.2_p14 | txt_ds_ps_ci_lower | -0.53 | -0.53 | TRUE |
| section_4.2_p14 | txt_ds_ps_ci_upper | 0.70 | 0.70 | TRUE |
| section_4.2_p14 | txt_ds_ps_wcb_high | 0.53 | 0.53 | TRUE |
| section_4.2_p14 | txt_ds_ps_wcb_low | -0.34 | -0.34 | TRUE |
| section_4.2_p14 | txt_ds_wcb_high | 0.57 | 0.57 | TRUE |
| section_4.2_p14 | txt_ds_wcb_low | -0.34 | -0.34 | TRUE |
| section_4.2_p14 | txt_naive_ate_coef | 0.126 | 0.126 | TRUE |
| section_4.2_p14 | txt_naive_ate_pval | 0.057 | 0.057 | TRUE |
| section_4.2_p14 | txt_naive_ate_se | 0.066 | 0.066 | TRUE |
| section_4.2_p14 | txt_no_ds_ci_lower | -1.67 | -1.67 | TRUE |
| section_4.2_p14 | txt_no_ds_ci_upper | 1.84 | 1.84 | TRUE |
| section_4.2_p14 | txt_no_ds_wcb_high | 1.71 | 1.71 | TRUE |
| section_4.2_p14 | txt_no_ds_wcb_low | -1.54 | -1.54 | TRUE |
| section_4.2_p14 | txt_reduction_pct | 0.65 | | FALSE |
| section_4.3_p15 | app_naive_sig_10 | | | |
| section_4.3_p15 | app_no_delta_star_05 | | | |
| section_4.3_p15 | txt_delta_star | 0.07 | | FALSE |
| Table 1 | t1_moderate_did_not_respond | 264 | 264 | TRUE |
| Table 1 | t1_moderate_invited | 50 | 50 | TRUE |
| Table 1 | t1_moderate_n_initial | 995 | 995 | TRUE |
| Table 1 | t1_moderate_responded | 731 | 731 | TRUE |
| Table 1 | t1_moderate_responded_to_invitation | 39 | 39 | TRUE |
| Table 1 | t1_polarized_did_not_respond | 272 | 272 | TRUE |
| Table 1 | t1_polarized_invited | 50 | 50 | TRUE |
| Table 1 | t1_polarized_n_initial | 985 | 985 | TRUE |
| Table 1 | t1_polarized_responded | 713 | 713 | TRUE |
| Table 1 | t1_polarized_responded_to_invitation | 33 | 33 | TRUE |
| Table 1 | t1_total_did_not_respond | 536 | 536 | TRUE |
| Table 1 | t1_total_invited | 100 | 100 | TRUE |
| Table 1 | t1_total_n_initial | 1980 | 1980 | TRUE |
| Table 1 | t1_total_responded | 1444 | 1444 | TRUE |
| Table 1 | t1_total_responded_to_invitation | 72 | 72 | TRUE |
| Table 2 | app_samples_similar | | | |
| Table 2 | t2_moderate_ds_mean | 3.583 | 3.583 | TRUE |
| Table 2 | t2_moderate_ds_n | 39 | 39 | TRUE |
| Table 2 | t2_moderate_ds_sd | 1.367 | 1.367 | TRUE |
| Table 2 | t2_moderate_ir_mean | 3.542 | 3.542 | TRUE |
| Table 2 | t2_moderate_ir_n | 731 | 731 | TRUE |
| Table 2 | t2_moderate_ir_sd | 1.243 | 1.243 | TRUE |
| Table 2 | t2_polarized_ds_mean | 3.826 | 3.826 | TRUE |
| Table 2 | t2_polarized_ds_n | 33 | 33 | TRUE |
| Table 2 | t2_polarized_ds_sd | 1.313 | 1.313 | TRUE |

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured. *(continued)*

| Location | Claim | Paper | Archive | Match |
|---|---|---|---|---|
| Table 2 | t2_polarized_ir_mean | 3.668 | 3.668 | TRUE |
| Table 2 | t2_polarized_ir_n | 713 | 713 | TRUE |
| Table 2 | t2_polarized_ir_sd | 1.270 | 1.270 | TRUE |
| Table 3 | app_ds_narrows | | | |
| Table 3 | app_wcb_uninformative | | | |
| Table 3 | t3_ds_ci_lower | -0.5283 | -0.5283 | TRUE |
| Table 3 | t3_ds_ci_upper | 0.7452 | 0.7452 | TRUE |
| Table 3 | t3_ds_ps_ci_lower | -0.5290 | -0.5290 | TRUE |
| Table 3 | t3_ds_ps_ci_upper | 0.6966 | 0.6966 | TRUE |
| Table 3 | t3_ds_ps_var_high | 0.0108 | 0.0108 | TRUE |
| Table 3 | t3_ds_ps_var_low | 0.0126 | 0.0126 | TRUE |
| Table 3 | t3_ds_ps_wcb_high | 0.5257 | 0.5257 | TRUE |
| Table 3 | t3_ds_ps_wcb_low | -0.3444 | -0.3444 | TRUE |
| Table 3 | t3_ds_var_high | 0.0111 | 0.0111 | TRUE |
| Table 3 | t3_ds_var_low | 0.0129 | 0.0129 | TRUE |
| Table 3 | t3_ds_wcb_high | 0.5718 | 0.5718 | TRUE |
| Table 3 | t3_ds_wcb_low | -0.3417 | -0.3417 | TRUE |
| Table 3 | t3_no_ds_ci_lower | -1.6691 | -1.6691 | TRUE |
| Table 3 | t3_no_ds_ci_upper | 1.8359 | 1.8359 | TRUE |
| Table 3 | t3_no_ds_var_high | 0.0059 | 0.0059 | TRUE |
| Table 3 | t3_no_ds_var_low | 0.0062 | 0.0062 | TRUE |
| Table 3 | t3_no_ds_wcb_high | 1.7097 | 1.7097 | TRUE |
| Table 3 | t3_no_ds_wcb_low | -1.5391 | -1.5391 | TRUE |

Of the 171 recorded claims, 131 can be compared against the deposited scripts. 128 match the published value and 3 are unreachable, all because of the stale object name.

## 7 Maintained rewrite

The rewrite lives in `maintained/`: nine scripts covering the three main text tables, the two appendix tables, both figures, the numbers Sections 4.2 and 4.3 state in prose, and the claim-by-claim audit. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used, and the `attrition` functions that implement the method (`estimator_ev`, `estimator_ds`, `sensitivity_ds`, and the internal `ds_manski_cis_2s` the simulation calls) are used unchanged, since they are the paper's contribution rather than incidental tooling.

Table 5: Scripts in the maintained rewrite.

| Script | Produces |
|---|---|
| helpers.R | packages, data, shared row labels |
| table_1_attrition.R | Table 1: attrition by condition |
| table_2_outcome_means.R | Table 2: outcome means by condition and response type |
| table_3_bounds_main.R | Table 3: bounds with and without double sampling |
| table_a1_mipo_effects.R | Appendix Table 1: effects assuming MIPO |
| table_a2_pairwise_bounds.R | Appendix Table 2: bounds for all pairwise comparisons |
| figure_1_simulation.R | Figure 1 and Appendix Figure 1: the simulation study |

| | |
|---|---|
| figure_2_sensitivity.R | Figure 2: the sensitivity analysis |
| text_section_4_claims.R | the prose numbers in Sections 4.2 and 4.3 |
| in_text_claims.R | one printed line per published claim, recomputed independently |

Two structural choices are worth stating. `figure_1_simulation.R` writes both Figure 1 and Appendix Figure 1, which differ only in whether the larger sample size is shown, because they come from one simulation grid and splitting them would mean running it twice. `text_section_4_claims.R` reads the CSV output of the table and figure scripts rather than refitting anything, so a number quoted in the text and the same number printed in a table cannot come apart.

## 7.1 Deprecated patterns replaced

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

| Original pattern | Replacement |
|---|---|
| ‘rm(list = ls())‘ | (omitted) |
| ‘setwd(”“)‘ | ‘here::here()‘ |
| ‘read.csv()‘ | ‘readr::read_csv()‘ |
| ‘lm()‘ + ‘commarobust::makerobustseslist()‘ | ‘estimatr::lm_robust(se_type = ”HC2”)‘ |
| ‘stargazer‘ + ‘out =‘ | ‘modelsummary(output = ...)‘ |
| ‘xtable‘ | ‘knitr::kable()‘ + ‘readr::write_lines()‘ |
| ‘multiplot()‘ (hand-rolled, grid-based) | ‘gridExtra::arrangeGrob()‘ |
| triple ‘for‘ loop building a matrix with ‘rbind‘ | ‘expand_grid()‘ + ‘pmap()‘ + ‘list_rbind()‘ |
| ‘reshape2::melt()‘ | ‘pivot_longer()‘ |
| ‘ifelse()‘ | ‘if_else()‘ |
| ‘data.frame()‘ | ‘tibble()‘ |
| ‘scale_color_grey(guide = FALSE)‘ | ‘scale_color_grey(guide = ”none”)‘ |
| ‘<ggplot> %+% data‘ | ‘<ggplot> + data‘ |
| magrittr pipe | native pipe |

`commarobust::commarobust()` is a wrapper around `lmtest::coeftest(fit, sandwich::vcovHC(fit, type = "HC2"))`, and `estimatr::lm_robust()` defaults to the same HC2 estimator, so the substitution is exact: the standard errors agree to fifteen digits. The one visible difference is in the significance stars. `commarobust::makerobustpslist()` overwrites the intercept's p-value with 1 before handing the models to `stargazer`, which is why the published Appendix Table 1 shows no stars on the constant. The rewrite reports the intercept's actual p-value, so `table_a1_mipo_effects.tex` stars the constant where the published table does not. No estimate or standard error differs.

The simulation deserves a note of its own. The archive passes four objects named `c1a_t`, `c1r_t`, `c2a_t` and `c2r_t` into `ds_manski_cis_2s()` without ever defining them. This does not error, because they map onto formal arguments the function body never touches and R does not evaluate an argument it does not use. They are vestigial cost parameters from an earlier version

of the function. The current function takes the arm-level moments as one list and has no such arguments; the rewrite builds that list directly.

# 8 Figures

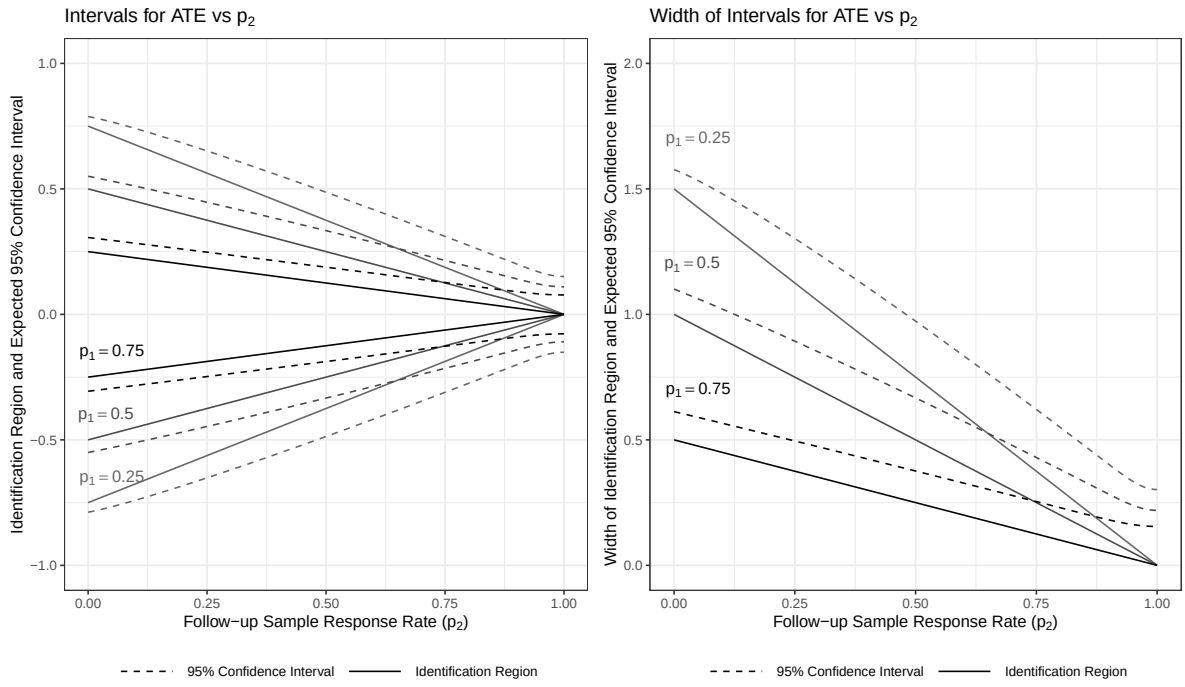



Figure 1: Figure 1 as reproduced by the maintained rewrite: the identification region and expected 95 percent confidence interval for the ATE against the follow-up response rate, at three initial response rates.

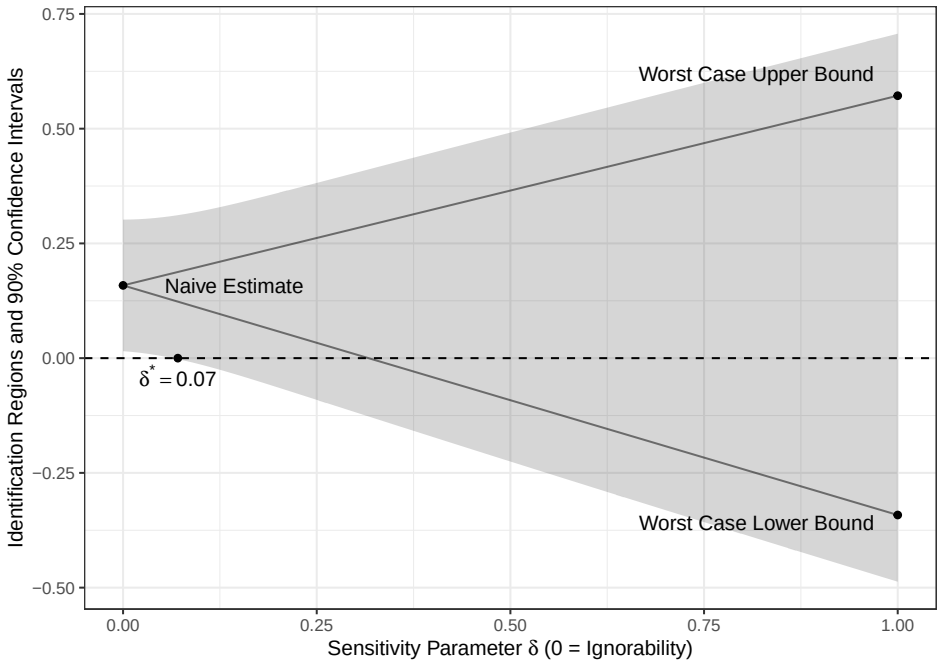



Figure 2: Figure 2 as reproduced by the maintained rewrite: the identification region and 90 percent confidence interval against the sensitivity parameter, with the delta star at which the interval first covers zero.

Neither figure carries printed estimates, so there is no panel label to check the way there is in some archives. Figure 2 is the exception that matters: it is annotated with $\delta^* = 0.07$, and that annotation is the only place the paper states the sensitivity result. The rewrite rebuilds both figures from its own data frames rather than saving the `ggplot` object `sensitivity_ds()` returns, so everything drawn here is produced by code in this repository. `figure_1_simulation.csv` carries the 6,000 point simulation grid behind Figure 1, which is what makes that figure diffable rather than merely inspectable.

## 9 Maintained rewrite verification

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number.

| Location | Claim | Paper | Rewrite | Match | Holds |
|---|---|---|---|---|---|
| Appendix Figure 1 | appx_ci_shrinks_region_does_not | | | | TRUE |
| Appendix Table 1 | ta1_all_ex_wave1_n | 2156 | 2156 | TRUE | |
| Appendix Table 1 | ta1_all_ex_wave1_r2 | 0.001 | 0.000969193379921518 | TRUE | |
| Appendix Table 1 | ta1_all_ex_wave2_n | 2156 | 2156 | TRUE | |
| Appendix Table 1 | ta1_all_ex_wave2_r2 | 0.0004 | 0.000359806687159292 | TRUE | |
| Appendix Table 1 | ta1_all_pp_wave1_n | 2156 | 2156 | TRUE | |

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

| Location | Claim | Paper | Rewrite | Match | Holds |
|---|---|---|---|---|---|
| Appendix Table 1 | ta1_all_pp_wave1_r2 | 0.024 | 0.0239507001082565 | TRUE | |
| Appendix Table 1 | ta1_all_pp_wave2_n | 2156 | 2156 | TRUE | |
| Appendix Table 1 | ta1_all_pp_wave2_r2 | 0.002 | 0.00174424183477495 | TRUE | |
| Appendix Table 1 | ta1_constant_ex_wave1_coef | 1.531 | 1.53146374829 | TRUE | |
| Appendix Table 1 | ta1_constant_ex_wave1_se | 0.024 | 0.0243596218569332 | TRUE | |
| Appendix Table 1 | ta1_constant_ex_wave2_coef | 1.617 | 1.61662106703145 | TRUE | |
| Appendix Table 1 | ta1_constant_ex_wave2_se | 0.024 | 0.024150796568631 | TRUE | |
| Appendix Table 1 | ta1_constant_pp_wave1_coef | 3.108 | 3.10841313269491 | TRUE | |
| Appendix Table 1 | ta1_constant_pp_wave1_se | 0.046 | 0.046462289784369 | TRUE | |
| Appendix Table 1 | ta1_constant_pp_wave2_coef | 3.542 | 3.54206566347466 | TRUE | |
| Appendix Table 1 | ta1_constant_pp_wave2_se | 0.046 | 0.0459614434759558 | TRUE | |
| Appendix Table 1 | ta1_placebo_ex_wave1_coef | 0.019 | 0.0190980494627954 | TRUE | |
| Appendix Table 1 | ta1_placebo_ex_wave1_se | 0.035 | 0.035282865017696 | TRUE | |
| Appendix Table 1 | ta1_placebo_ex_wave2_coef | -0.011 | -0.010581741188767 | TRUE | |
| Appendix Table 1 | ta1_placebo_ex_wave2_se | 0.034 | 0.0343376300527564 | TRUE | |
| Appendix Table 1 | ta1_placebo_pp_wave1_coef | 0.352 | 0.352378065806935 | TRUE | |
| Appendix Table 1 | ta1_placebo_pp_wave1_se | 0.067 | 0.067434267815023 | TRUE | |
| Appendix Table 1 | ta1_placebo_pp_wave2_coef | 0.042 | 0.041501752255645 | TRUE | |
| Appendix Table 1 | ta1_placebo_pp_wave2_se | 0.066 | 0.0656684123731185 | TRUE | |
| Appendix Table 1 | ta1_polarized_ex_wave1_coef | -0.031 | -0.0311131171539694 | TRUE | |
| Appendix Table 1 | ta1_polarized_ex_wave1_se | 0.034 | 0.034340859741032 | TRUE | |
| Appendix Table 1 | ta1_polarized_ex_wave2_coef | -0.030 | -0.0296645452923338 | TRUE | |
| Appendix Table 1 | ta1_polarized_ex_wave2_se | 0.034 | 0.03402111975226 | TRUE | |
| Appendix Table 1 | ta1_polarized_pp_wave1_coef | 0.468 | 0.468375086098889 | TRUE | |
| Appendix Table 1 | ta1_polarized_pp_wave1_se | 0.067 | 0.0665244501080027 | TRUE | |
| Appendix Table 1 | ta1_polarized_pp_wave2_coef | 0.126 | 0.125886650690804 | TRUE | |
| Appendix Table 1 | ta1_polarized_pp_wave2_se | 0.066 | 0.0661527954807814 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_ci_lower | -0.4778 | -0.477842687179526 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_ci_upper | 0.5769 | 0.57692983337372 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_var_high | 0.0067 | 0.00666486134438053 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_var_low | 0.0083 | 0.0082562481709875 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_wcb_high | 0.4426 | 0.442646282695529 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_ex_wcb_low | -0.3284 | -0.328384873083365 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_ci_lower | -0.6119 | -0.611862038531696 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_ci_upper | 0.5044 | 0.504351636768822 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_var_high | 0.0114 | 0.0114143454384199 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_var_low | 0.0106 | 0.0106127353002281 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_wcb_high | 0.3286 | 0.328618863548512 | TRUE | |
| Appendix Table 2 | ta2_mod_vs_placebo_pp_wcb_low | -0.4424 | -0.442412292230382 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_ci_lower | -0.6087 | -0.608731543014259 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_ci_upper | 0.6032 | 0.603194110577457 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_var_high | 0.0095 | 0.00953316194586098 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_var_low | 0.0070 | 0.00701487544541756 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_wcb_high | 0.4426 | 0.442594036170701 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_ex_wcb_low | -0.4710 | -0.470967068846773 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_ci_lower | -0.5283 | -0.52830967410789 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_ci_upper | 0.7452 | 0.745174826433894 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_var_high | 0.0111 | 0.0111080739914738 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_var_low | 0.0129 | 0.0128647858310974 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_wcb_high | 0.5718 | 0.571815728388134 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_mod_pp_wcb_low | -0.3417 | -0.34174537662934 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_ci_lower | -0.6004 | -0.600425131607832 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_ci_upper | 0.6944 | 0.694385360492085 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_var_high | 0.0094 | 0.00938841779116446 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_var_low | 0.0085 | 0.00846151811732802 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_wcb_high | 0.5350 | 0.535009163087335 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_ex_wcb_low | -0.4491 | -0.449120786151244 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_ci_lower | -0.6171 | -0.617118472710443 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_ci_upper | 0.7262 | 0.726201234202598 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_var_high | 0.0114 | 0.0114487996375541 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_var_low | 0.0124 | 0.012403901338986 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_wcb_high | 0.5502 | 0.550203436157751 | TRUE | |
| Appendix Table 2 | ta2_pol_vs_placebo_pp_wcb_low | -0.4339 | -0.433926513080827 | TRUE | |
| appendix_section_1_p2 | appx_n_large | 80000 | 80000 | TRUE | |

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

| Location | Claim | Paper | Rewrite | Match | Holds |
|---|---|---|---|---|---|
| appendix_section_1_p2 | appx_n_small | 800 | 800 | TRUE | |
| appendix_section_1_p2 | appx_n2_large | 5000 | 5000 | TRUE | |
| appendix_section_1_p2 | appx_n2_small | 50 | 50 | TRUE | |
| appendix_section_1_p2 | appx_size_factor | 100 | 100 | TRUE | |
| appendix_section_1_p2 | appx_two_scenarios | 2 | 2 | TRUE | |
| appendix_section_2_p3 | appx_effect_diminished | | | | TRUE |
| appendix_section_2_p3 | appx_extremity_negligible | | | | TRUE |
| appendix_section_2_p3 | appx_wave1_effect | 0.468 | 0.468375086098889 | TRUE | |
| appendix_section_2_p3 | appx_wave1_effect_strong | | | | TRUE |
| appendix_section_2_p3 | appx_wave2_effect | 0.126 | 0.125886650690804 | TRUE | |
| appendix_section_3_p4 | appx_pairwise_columns | 6 | 6 | TRUE | |
| Figure 1 | sim_p2_zero_is_wcb | | | | TRUE |
| Figure 1 | sim_point_identified | | | | TRUE |
| Figure 1 | sim_region_smaller_in_p1 | | | | TRUE |
| Figure 1 | sim_width_shrinks_to_zero | | | | TRUE |
| Figure 2 | fig2_delta_star | 0.07 | 0.0707070707070707 | TRUE | |
| section_3_p11 | sim_n_followup | 50 | 50 | TRUE | |
| section_3_p11 | sim_n_total | 800 | 800 | TRUE | |
| section_3_p11 | sim_p1_max | 0.75 | 0.75 | TRUE | |
| section_3_p11 | sim_p1_min | 0.25 | 0.25 | TRUE | |
| section_3_p11 | sim_p2_max | 1 | 1 | TRUE | |
| section_3_p11 | sim_p2_min | 0 | 0 | TRUE | |
| section_3_p11 | sim_true_ate | 0 | 0 | TRUE | |
| section_3_p12 | sim_n_scenarios | 3 | 3 | TRUE | |
| section_3_p12 | sim_p1_scenario_high | 0.75 | 0.75 | TRUE | |
| section_3_p12 | sim_p1_scenario_low | 0.25 | 0.25 | TRUE | |
| section_3_p12 | sim_p1_scenario_mid | 0.50 | 0.5 | TRUE | |
| section_4.1_p14 | app_n_completed | 72 | 72 | TRUE | |
| section_4.1_p14 | app_n_invited | 100 | 100 | TRUE | |
| section_4.1_p14 | app_n_invited_per_arm | 50 | 50 | TRUE | |
| section_4.1_p14 | app_n_responded | 1444 | 1444 | TRUE | |
| section_4.1_p14 | app_n_subjects | 1980 | 1980 | TRUE | |
| section_4.2_p14 | app_outcome_max | 6 | 6 | TRUE | |
| section_4.2_p14 | app_outcome_min | 0 | -1.55889440680195e-15 | TRUE | |
| section_4.2_p14 | txt_ci_width_ds_ps | 1.23 | 1.22561085834396 | TRUE | |
| section_4.2_p14 | txt_ci_width_no_ds | 3.50 | 3.50496018440374 | TRUE | |
| section_4.2_p14 | txt_ds_ps_ci_lower | -0.53 | -0.529010551320638 | TRUE | |
| section_4.2_p14 | txt_ds_ps_ci_upper | 0.70 | 0.696600307023322 | TRUE | |
| section_4.2_p14 | txt_ds_ps_wcb_high | 0.53 | 0.525683688852529 | TRUE | |
| section_4.2_p14 | txt_ds_ps_wcb_low | -0.34 | -0.344389910863888 | TRUE | |
| section_4.2_p14 | txt_ds_wcb_high | 0.57 | 0.571815728388134 | TRUE | |
| section_4.2_p14 | txt_ds_wcb_low | -0.34 | -0.34174537662934 | TRUE | |
| section_4.2_p14 | txt_naive_ate_coef | 0.126 | 0.125886650690804 | TRUE | |
| section_4.2_p14 | txt_naive_ate_pval | 0.057 | 0.0571779950234041 | TRUE | |
| section_4.2_p14 | txt_naive_ate_se | 0.066 | 0.0661527954807814 | TRUE | |
| section_4.2_p14 | txt_no_ds_ci_lower | -1.67 | -1.66907903885775 | TRUE | |
| section_4.2_p14 | txt_no_ds_ci_upper | 1.84 | 1.83588114554598 | TRUE | |
| section_4.2_p14 | txt_no_ds_wcb_high | 1.71 | 1.70966762747749 | TRUE | |
| section_4.2_p14 | txt_no_ds_wcb_low | -1.54 | -1.53914496339566 | TRUE | |
| section_4.2_p14 | txt_reduction_pct | 0.65 | 0.650321032519101 | TRUE | |
| section_4.3_p15 | app_naive_sig_10 | | | | TRUE |
| section_4.3_p15 | app_no_delta_star_05 | | | | TRUE |
| section_4.3_p15 | txt_delta_star | 0.07 | 0.0707070707070707 | TRUE | |
| Table 1 | t1_moderate_did_not_respond | 264 | 264 | TRUE | |
| Table 1 | t1_moderate_invited | 50 | 50 | TRUE | |
| Table 1 | t1_moderate_n_initial | 995 | 995 | TRUE | |
| Table 1 | t1_moderate_responded | 731 | 731 | TRUE | |
| Table 1 | t1_moderate_responded_to_invitation | 39 | 39 | TRUE | |
| Table 1 | t1_polarized_did_not_respond | 272 | 272 | TRUE | |
| Table 1 | t1_polarized_invited | 50 | 50 | TRUE | |
| Table 1 | t1_polarized_n_initial | 985 | 985 | TRUE | |
| Table 1 | t1_polarized_responded | 713 | 713 | TRUE | |
| Table 1 | t1_polarized_responded_to_invitation | 33 | 33 | TRUE | |
| Table 1 | t1_total_did_not_respond | 536 | 536 | TRUE | |

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

| Location | Claim | Paper | Rewrite | Match | Holds |
|---|---|---|---|---|---|
| Table 1 | t1_total_invited | 100 | 100 | TRUE | |
| Table 1 | t1_total_n_initial | 1980 | 1980 | TRUE | |
| Table 1 | t1_total_responded | 1444 | 1444 | TRUE | |
| Table 1 | t1_total_responded_to_invitation | 72 | 72 | TRUE | |
| Table 2 | app_samples_similar | | | | TRUE |
| Table 2 | t2_moderate_ds_mean | 3.583 | 3.58333333333333 | TRUE | |
| Table 2 | t2_moderate_ds_n | 39 | 39 | TRUE | |
| Table 2 | t2_moderate_ds_sd | 1.367 | 1.36730301830269 | TRUE | |
| Table 2 | t2_moderate_ir_mean | 3.542 | 3.54206566347469 | TRUE | |
| Table 2 | t2_moderate_ir_n | 731 | 731 | TRUE | |
| Table 2 | t2_moderate_ir_sd | 1.243 | 1.2426600835923 | TRUE | |
| Table 2 | t2_polarized_ds_mean | 3.826 | 3.82575757575758 | TRUE | |
| Table 2 | t2_polarized_ds_n | 33 | 33 | TRUE | |
| Table 2 | t2_polarized_ds_sd | 1.313 | 1.31322130252061 | TRUE | |
| Table 2 | t2_polarized_ir_mean | 3.668 | 3.6679523141655 | TRUE | |
| Table 2 | t2_polarized_ir_n | 713 | 713 | TRUE | |
| Table 2 | t2_polarized_ir_sd | 1.270 | 1.27045080160387 | TRUE | |
| Table 3 | app_ds_narrows | | | | TRUE |
| Table 3 | app_wcb_uninformative | | | | TRUE |
| Table 3 | t3_ds_ci_lower | -0.5283 | -0.52830967410789 | TRUE | |
| Table 3 | t3_ds_ci_upper | 0.7452 | 0.745174826433894 | TRUE | |
| Table 3 | t3_ds_ps_ci_lower | -0.5290 | -0.529010551320638 | TRUE | |
| Table 3 | t3_ds_ps_ci_upper | 0.6966 | 0.696600307023322 | TRUE | |
| Table 3 | t3_ds_ps_var_high | 0.0108 | 0.0107972726598501 | TRUE | |
| Table 3 | t3_ds_ps_var_low | 0.0126 | 0.0125981273119329 | TRUE | |
| Table 3 | t3_ds_ps_wcb_high | 0.5257 | 0.525683688852529 | TRUE | |
| Table 3 | t3_ds_ps_wcb_low | -0.3444 | -0.344389910863888 | TRUE | |
| Table 3 | t3_ds_var_high | 0.0111 | 0.0111080739914738 | TRUE | |
| Table 3 | t3_ds_var_low | 0.0129 | 0.0128647858310974 | TRUE | |
| Table 3 | t3_ds_wcb_high | 0.5718 | 0.571815728388134 | TRUE | |
| Table 3 | t3_ds_wcb_low | -0.3417 | -0.34174537662934 | TRUE | |
| Table 3 | t3_no_ds_ci_lower | -1.6691 | -1.66907903885775 | TRUE | |
| Table 3 | t3_no_ds_ci_upper | 1.8359 | 1.83588114554598 | TRUE | |
| Table 3 | t3_no_ds_var_high | 0.0059 | 0.00588785669626298 | TRUE | |
| Table 3 | t3_no_ds_var_low | 0.0062 | 0.0062401008346893 | TRUE | |
| Table 3 | t3_no_ds_wcb_high | 1.7097 | 1.70966762747749 | TRUE | |
| Table 3 | t3_no_ds_wcb_low | -1.5391 | -1.53914496339566 | TRUE | |

Every comparable row carries `match_rewrite = TRUE`: **158** values produced by the maintained rewrite, all matching the published paper to the precision the page prints, and **13** descriptive claims that hold. Every value here is read out of a file in `maintained/output/` by `build_ground_truth.R` rather than transcribed, so a value in this table that disagreed with the pipeline would be a failure of the pipeline and not of the transcription.

# 10 R environment

| Item | Value |
|---|---|
| R version | 4.6.2 |
| Platform | aarch64-apple-darwin23 |

| Date run | 2026-11-12 |
|---|---|

Table 9: Package versions used for the run behind this report.

| Package | Version |
|---|---|
| attrition | 1.0.0 |
| estimatr | 2.0.1 |
| modelsummary | 2.6.0 |
| dplyr | 1.2.1 |
| ggplot2 | 4.0.3 |
| tidyr | 1.3.2 |
| purrr | 1.2.2 |
| readr | 2.2.0 |
| here | 1.0.2 |
| gridExtra | 2.3.1 |
| knitr | 1.52 |
| kableExtra | 1.4.1 |

Table 10: The deposited files, as pinned in the manifest. All seven served checksums match the checksums Dataverse publishes, and all seven match the local copy this work was done against.

| File | Bytes | MD5 (served) | Published MD5 agrees |
|---|---|---|---|
| README.txt | 2090 | ea327d0299d36ebd0f5bc1b4a01e5c6d | yes |
| attrition_0.0.0.9000.tar.gz | 34511 | 7fc29b91569bff874735191147a65fc8 | yes |
| commarobust_0.1.0.tar.gz | 8085 | feef406b22d3259c23d16f91ffbaadc9 | yes |
| experimental_analysis_appendix.R | 3756 | 5fc6e7d1fe25133820d96648ddb7ebfb | yes |
| experimental_analysis_main_text.R | 3668 | 19a3cf55e8685f65d23419cd013fae0e | yes |
| levendusky_mturk_clean.csv | 178043 | 6290d4802c5331dfa550fc954ccf07d0 | yes |
| simulation_analysis.R | 6358 | ebdc8595bddfdef33ee8dde128366039 | yes |